

Gaurav Vanvari

Software Engineer

Odenton, MD

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Github: @gvanvari

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Portfolio: <https://gvanvari.github.io/>

Software Engineer with 8 years of experience in functional, load, chaos, LLM, and security testing, seeking to leverage expertise in a Security Engineer role.

Professional Experience

Bloomberg LP | Princeton, NJ

Sep 2020 - Present

Bloomberg's EMS manages end-to-end workflows, enabling dynamic control of positions and portfolios

Software Engineer

- API Design: Designed and developed a Python-based backend testing application and APIs, leveraging BDD, Behave, Docker, Kafka, and Jenkins to automate 90% of EMSX workflows, and implemented shift-left testing practices to detect defects earlier in the development lifecycle
- FIX Protocol Testing: Developed the infrastructure to test FIX workflows using QuickFix and Python
- CI/CD Automation: Automated CI/CD pipelines with integrated documentation publishing (Alexandria, Sphinx), secrets detection, linting, and unit test coverage to ensure secure and quality deployments
- Observability: Integrated distributed tracing enabling end-to-end request observability, automated performance anomaly detection, and continuous security monitoring in line with NIST SP 800-137 best practices
- Regulatory Compliance: Advanced EMSX's Digital Operational Resilience Act (DORA) compliance by enhancing access control lists, mitigating SQL injection risks, and migrating components to appsh
- UI Test Automation: Automated 90% of UI workflows using TypeScript, BDD, and Cucumber, reducing regression testing time by 40% and supporting CI/CD via daily Jenkins builds on Windows and Linux in the Bloomberg Cloud
- Tool Development: Developed a scenario management and status reporting tool with TypeScript, R+, and Comdb2 to improve test coverage, classify test failures, and support shift-left initiatives
- Machine Learning: Automated ticket creation with Python and scikit-learn, leveraging AWS S3 error logs to reduce triage time and manual intervention
- Project Leadership: Acted as Scrum Master, leading knowledge sharing, fostering product awareness, and supporting team growth through trainings and demos
- Cross-Team Initiatives: Co-led GitHub Copilot Working Group, formalizing secure, AI-driven development and testing workflows, and promoting automated code generation in line with organizational best practices

IronNet Cybersecurity | Tysons Corner, VA

Sep 2017 - July 2020

IronDefense, IronNet's Next-Generation AI/ML Network Traffic Analysis (NTA) solution, detects network anomalies using machine learning and threat intelligence along the Cyber Kill Chain.

Software Engineer in Test

- Development: Developed and tested IronDefense API's and integrations with various tools (CrowdStrike, Phantom, QRadar, ServiceNow, Splunk, Swimlane, and XSOAR) to expedite security alert triage using Go, Python, Typescript, and Docker for incident response in line with NIST SP 800-171 best practices
- Web Application Security: Collaborated with cross-functional teams to get IronDefense FedRAMP certified by configuring Snyk via GitHub Actions workflows in repositories, enabling the engineering team to identify and fix security vulnerabilities and shift left
- Machine Learning Model Validation: Generated network anomalies using Ixia to simulate cyber attacks, enabling data scientists to prototype and deploy ML threat detection models on-premise servers

- Threat Hunting: Tested Hunt Platform and Splunk queries in collaboration with Threat Hunters
- UI Test Automation: Automated 15% of manual tests through Typescript, Cypress and TeamCity, boosting regression test runs tenfold and optimizing efficiency
- Backend Test Automation: Automated functional and REST API testing by replacing TestRail test cases with Go, gRPC, and Postman for nightly TeamCity runs
- Collaboration: Authored sales, customer, and SRE documentation, ensuring a seamless deployment process, and supported the Sales team in delivering effective IronDefense demonstrations to partners

Cogent Infotech | Salem, OR

May 2017 – Sep 2017

Software Engineer

- Developed Java-based batch jobs using HQL and Hibernate to automate processes like Assistance Status Change and display results for workers and case managers

MediaTek USA Inc. | San Jose, CA

May 2015 - Aug 2015

Software Engineering Intern

- Developed a C++ packet generator and an efficient CPU-to-GPU queue following Heterogeneous System Architecture and OpenCL 2.0 specifications to optimize GPU processing

Certification and Training

API Security Foundations and Securing LLM & NLP APIs, APIsec University

May 2025

OWASP API Security Top 10, Prompt Injection

Security+, CompTIA

Dec 2023

App Vulnerabilities & Attacks, Risk Management, Protecting Backups, Cryptography, Hashing, Data Classification

Jr Pentester, TryHackMe

Oct 2023

OWASP Top 10, CVE, CVSS, Burp Suite, nmap, Metasploit, John the Ripper, Gobuster, Hydra, Msfvenom

Certified Cloud Practitioner, AWS

Jan 2020

Availability Zones, AWS CloudTrail, Amazon EC2, AWS Billing Console, Amazon S3, Route53, IAM

Academic Projects

Multimodal Chatbot

Dec 2025

Developed a multi-agent conversation system with human in loop that simulates budget discussions between political, citizen, and treasury representatives using AI agents powered by OpenAI's LLM, Gradio, Python and autogen

Workout Tracker

Dec 2025

A web application that allows users to track their workouts and monitor their progress over time through an intuitive interface using Typescript, Node, Express and SQLite

FPGA vs GPU performance comparison for Digit Recognition, NYU

Spring 2016

Evaluated hardware acceleration for convolution layers in a machine learning neural network for digit recognition (SVHN dataset, 27,000 entries), implementing models in VHDL (Spartan 6 FPGA) and LUA (NVIDIA Titan GPU), and found the FPGA achieved an 800x speedup over the GPU

Advanced Encryption Standard, NYU

Fall 2015

Enhanced AES-256 implementation on Spartan-6 by leveraging on-chip Digital Clock Manager and Block RAM for efficient key storage and clock generation, while integrating an authentication module to allow only authorized key modifications—improving both speed and security

RC5 Encryption Algorithm, NYU

Fall 2014

Implemented RC5 block cipher with 64-bit input, 12 rounds, a 16-byte user key, and an accompanying RC5 Round Key Generation module with 26 words and 12 rounds using VHDL and Spartan-6 FPGA

Education

New York University | Brooklyn, NY

May 2016

Master of Science in Computer Engineering, GPA: 3.4/4

Indraprastha University (GGSIPU) | New Delhi, India

June 2014

B.Tech in Electronics and Communications Engineering, GPA 3.5/4.0